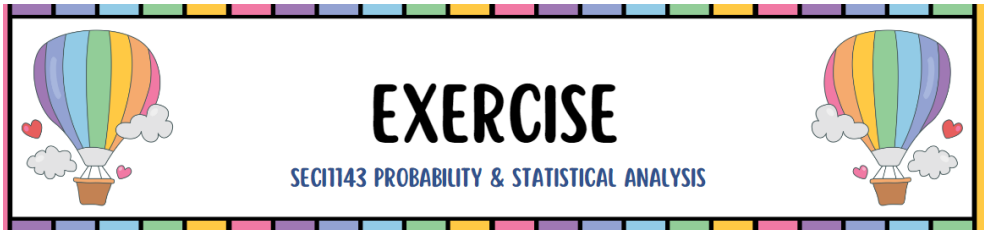
FACULTY OF COMPUTING
UTM Johor Bahru



CHAPTER 1

Question 1

Complete Table 1 by identifying data type and levels of measurement:
- Data Type (qualitative, quantitative)
- Level of Measurement (nominal, ordinal, interval, ratio)

No	Variable	Data Type	Level of Measurement
1.	The time that a student spends to complete an assignment.		
2.	Your gender status: Male/Female		
3.	The winning time for a horse running in a competition.		
4.	A patient expresses the amount of pain he/she feels on a 10-point scale where 1 is no pain and 10 means extreme pain.		
5.	How high do kangaroos jump?		

Question 2

Identify the appropriate answers of four levels of measurements (nominal, ordinal, interval, and ratio).

a) How much time do you spend on self-learning daily with the options less than 2 hours, 3 to 4 hours, and more than 5 hours?

b) The CGPA of students in the semester 2 can be categorized into 3.00<=CGPA<3.33, 3.33<=CGPA<3.67, and 3.67<=CGPA<=4.00.

c) The number of elections for ABC and XYZ in town K has been voted, and its results are 1023 and 2143.

d) How could you express your feeling of pain now? Moderate, strong, or intense?

e) How likely is it that you would recommend this JKL company to your friend or others? Kindly give your rate in a range of 0 to 5.

Mid-Term 2018/2019

Question 1 **[10 MARKS]**

Medical records of cancer patients in Hospital ABC for one month are given in Table 1 below.

Table 1

	Gender (M/F)	Age	Weight (kg)	Height (cm)	Smoking (1=No, 2=Yes)
Patient 1	M	59	75	169	1
Patient 2	M	67	59	152	2
Patient 3	F	73	45	143	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
Patient 100	F	45	66	155	1

According to Table 1, identify the following:

(a) Population (1 mark)
(b) Sample (1 mark)
(c) Is it primary or secondary data? Why? (2 marks)
(d) Which variable(s) is/are quantitative? (1½ marks)
(e) Which variable(s) is/are qualitative? (1½ marks)
(f) Nominal level of measurement variable(s) (1½ marks)
(g) Ratio level of measurement variable(s) (1½ marks)

Mid-Term 2017/2018

QUESTION 1 **(10 MARKS)**

a) Medical records of four of the patients from a total of 30 patients of diabetes are listed in Table 1.

Table 1

Name	Age	Gender	Blood Sugar
Janet	30	Female	8.5 mol/L
Jonas	35	Male	8.0 mol/L
Steven	33	Male	7.9 mol/L
Zack	24	Male	9.0 mol/L

According to Table 1, identify the following:

i) Quantitative data [0.5 Mark]
ii) Qualitative data [0.5 Mark]
iii) Sample [0.5 Mark]
iv) Population [0.5 Mark]

b) List and explains the four levels of measurements and provide ONE example for each level of measurements. [2 Marks]

c) Determine the following data whether discrete or continuous.

i) Height of 35 students in a class. [1 Mark]
ii) Number of cars entered the recreation park in weekend. [1 Mark]
iii) The number of fresh graduates from UTM in 2018. [1 Mark]
iv) Average speed of cars passing the federal highway. [1 Mark]

d) List TWO examples of both primary and secondary data sources. [2 Marks]

Mid-Term 2016/2017

Question 1 [10 marks]

a) A survey will be given to 100 students randomly selected from the freshmen class at Lincoln High School.

i) What is the population? (1 mark)

ii) What is the sample? (1 mark)

b) State the scale/level of measurement for the following:

i) City of birth (1 mark)

ii) A score on a 5-point quiz measuring knowledge of algebra. (1 mark)

iii) Comparing the number of girls to the number of boys in a classroom. (1 mark)

Mid-Term 2015/2016

Question 1 **(5 marks)**

Determine which of the four levels of measurement (nominal, ordinal, interval and ratio) is most appropriate. [5 marks]

a. Heights of women basketball players in a tournament.
b. Ratings of superior, above average, average, below average or poor for blind dates.
c. Noon temperatures (in degrees Celsius) in a Johor Bahru this week.
d. A movie critic's classification of "drama, comedy, adventure".
e. Consumer Reports magazine ratings of "best buy, recommended, not recommended".
f. Distances travelled by students who commute to faculty.
g. The actual contents (in ml) of cola in Pepsi cans.
h. IQ scores, where the score is considered to be a measure of intelligence.
i. IQ scores, where the score is considered to be the number of points scored on the IQ test.
j. The number of bugs made when a programmer develop a coding for a project.